

Econ 272 Department of Economics
Spring 2024

GSB 607
Stanford University

PROBLEM SET VII

DUE: WEDNESDAY JUNE 4th, 2025

In this problem set you have to use the California smoking data from the Abadie et al synthetic control paper. Let $t = 1, \dots, T_0$ denote the preperiods and $t = T_0 + 1, \dots, T$ denote the post periods. There are data for N states, with unit N denoting California.

1. Estimate the effect of the legislation on smoking rates in California, for each year post legislation using a simple difference in differences estimator, using only the post-legislation data from the period you are estimating the effect for.
2. Estimate the effect for the first post legislation year $T_0 + 1$ using the Abadie et al synthetic control method, using their specification. You can use their synthetic control code.
3. Consider the regression

$$Y_{N,t} = \sum_{i=1}^{N-1} \gamma_i \cdot Y_{i,t},$$

to be estimated on the T_0 pre periods, where state N corresponds to California. Using only the first ten control states to avoid perfect collinearity, estimate γ by least squares without any restrictions (neither the adding up restriction, nor the nonnegativity restriction.) Predict Y_{N,T_0+1} . How does the treatment effect compare to the DID and SC estimates?

4. Estimate the same regression using all control states, using the minimum L_2 estimate to deal with nonuniqueness.
5. Consider the regression

$$Y_{i,T_0+1} = \sum_{t=1}^{T_0} \delta_t \cdot Y_{i,t},$$

to be estimated on the states other than California, and use that to predict Y_{N,T_0+1} for California, again using the minimum L_2 norm if there are problems with non-uniqueness. Do not impose any restrictions on the parameters. How does the estimate for the treatment effect compare to the other estimates.

6. Estimate the treatment effect on Texas, using both the DID estimator and the SC estimator (with nonnegativity and adding up restrictions), on the first post-treatment year. How do the magnitudes compare to the estimates for California?